

Lucas Nunes

POSTDOCTORAL RESEARCHER · RWTH AACHEN UNIVERSITY

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Research Interests

Generative models · Representation learning · 3D scene understanding · Perception

Education

Dr.-Ing. (Ph.D.) Degree - *summa cum laude* / with distinction

University of Bonn

THESIS: "LEARNING DISCRIMINATIVE REPRESENTATIONS AND GENERATIVE APPROACHES FOR OUTDOOR 3D LIDAR DATA"

Feb/2026

(SUPERVISOR: PROF. DR. CYRILL STACHNISS)

M.Sc. in Computer Science

University of São Paulo

THESIS: "ENVIRONMENT RECONSTRUCTION ON DISPARITY IMAGES USING SURFACE FEATURES AND GENERATIVE ADVERSARIAL NETWORKS"

Mar/2020

(SUPERVISOR: PROF. DR. DENIS F. WOLF)

Bachelor in Computer Science

University of São Paulo

DEGREE IN COMPUTER SCIENCE

Jan/2018

Academic Positions

Postdoctoral Researcher

RWTH Aachen University

COMPUTER VISION GROUP

Mar/2026 - Present

Ph.D. Student and Research Assistant

University of Bonn

PHOTOGRAMMETRY & ROBOTICS LAB, INSTITUTE OF GEODESY AND GEOINFORMATION

Nov/2020 - Dec/2025

Intern Researcher

Karlsruhe Institute of Technology

INSTITUT FÜR MESS- UND REGELUNGSTECHNIK

Aug/2018 - Nov/2018

Master Student and Research Assistant

University of São Paulo

MOBILE ROBOTICS LABORATORY, INSTITUTE OF MATHEMATICS AND COMPUTER SCIENCE

Feb/2018 - Mar/2020

Junior Research Assistant

University of São Paulo

MOBILE ROBOTICS LABORATORY, INSTITUTE OF MATHEMATICS AND COMPUTER SCIENCE

Ago/2014 - Jun/2017

Awards

Faculty Award 2026 for Geodesy for *Towards Generating Realistic 3D Semantic Training Data for Autonomous Driving* (L. Nunes et al., PAMI)

2026

Finalist - IROS 2025 Best Agri-Robotics Paper Award for *Zero-Shot Semantic Segmentation for Robots in Agriculture*

2025

Teaching

Mobile Sensing and Robotics I - University of Bonn, MSc, 2h/week, shared teaching, winter term

2025

Mobile Sensing and Robotics II - University of Bonn, MSc, 2h/week, lecture, summer term

2025

Mobile Sensing and Robotics II - University of Bonn, MSc, 2h/week, shared teaching, summer term

2024

Techniques for Self-Driving Cars - University of Bonn, MSc, 2h/week, single lecture, winter term

2023

Open-World Panoptic Segmentation of Traffic Participants - University of Bonn, MSc, project, every term

2023

Machine Learning for Robotics and Computer Vision - University of Bonn, MSc, 2h/week, tutoring, summer term

2022

Machine Learning for Robotics and Computer Vision - University of Bonn, MSc, 2h/week, tutoring, summer term

2021

Online Teaching Examples (Links to Youtube Videos) ---

Self-Driving Cars: Unsupervised Learning for Vehicles Perception

Link: <https://www.youtube.com/watch?v=9KA04ayP2P4>

2023

Supervision ---

Novel View Synthesis of Indoor Dynamic Scenes

University of Zurich, MSc thesis

2024

Fruits 3D Reconstruction for Agricultural Robotics with Diffusion Models

University of Bonn, MSc thesis

2025

Invited Talks ---

Data Science Sanca Meetup. São Carlos, Brazil on *How Autonomous Vehicles can perceive what is happening around and make decisions?*

Mar/2018

4CORP Meetup, Futurecom. São Paulo, Brazil. on *The Autonomous Vehicles Revolution: A New Relation Between User and Vehicle*

Out/2019

Academic Services ---

REVIEWING SERVICES

Reviewer for conferences: IROS, ICRA, IV, ICCV, NeurIPS, CVPR, ECCV, RSS

Reviewer for Journals: IEEE Robotics and Automation Letters, IEEE Transactions on Robotics, IEEE Transactions on Pattern Analysis and Machine Intelligence

WORKSHOP ORGANIZATION

ICRA Workshop on *Uncertainty in Open-World Robotics*

2026

Professional Experience ---

Software Engineer

DEVELOPMENT OF FRONT AND BACK-END FOR SAFETY ASSESSMENT TOOL AND URBAN SIMULATED ENVIRONMENTS IN THE CONTEXT OF AUTONOMOUS VEHICLES

IVEX
Apr/2020 – Nov/2020

Technical Leader

COORDINATION OF A WEB CRAWLER PROGRAMMING TEAM FOR DATA COLLECTION AND DEVELOPMENT/MAINTENANCE OF A BACK-END DATABASE SERVER

Juristec+
Jan/2019 – Apr/2020

Python Developer

DEVELOPMENT OF WEB CRAWLER PYTHON SCRIPTS FOR DATA COLLECTION

Juristec+
Jan/2018 – Aug/2018

Publication List

Peer-reviewed journal articles

- [1] L. Nunes, R. Marcuzzi, J. Behley, and C. Stachniss, “Towards Generating Realistic 3D Semantic Training Data for Autonomous Driving,” *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2026.
- [2] R. Marcuzzi, L. Nunes, E. Marks, L. Wiesmann, T. Labe, J. Behley, and C. Stachniss, “SfmOcc: Vision-Based 3D Semantic Occupancy Prediction in Urban Environments,” *IEEE Robotics and Automation Letters (RA-L)*, 2025.
- [3] L. Wiesmann, T. Labe, L. Nunes, J. Behley, and C. Stachniss, “Joint Intrinsic and Extrinsic Calibration of Perception Systems Utilizing a Calibration Environment,” *IEEE Robotics and Automation Letters (RA-L)*, vol. 9, no. 10, pp. 9103–9110, 2024.
- [4] R. Marcuzzi, L. Nunes, L. Wiesmann, J. Behley, and C. Stachniss, “Mask-Based Panoptic LiDAR Segmentation for Autonomous Driving,” *IEEE Robotics and Automation Letters (RA-L)*, vol. 8, no. 2, pp. 1141–1148, 2023.
- [5] R. Marcuzzi, L. Nunes, L. Wiesmann, E. Marks, J. Behley, and C. Stachniss, “Mask4D: End-to-End Mask-Based 4D Panoptic Segmentation for LiDAR Sequences,” *IEEE Robotics and Automation Letters (RA-L)*, vol. 8, no. 11, pp. 7487–7494, 2023.
- [6] L. Wiesmann, L. Nunes, J. Behley, and C. Stachniss, “KPPR: Exploiting Momentum Contrast for Point Cloud-Based Place Recognition,” *IEEE Robotics and Automation Letters (RA-L)*, vol. 8, no. 2, pp. 592–599, 2023.
- [7] X. Chen, B. Mersch, L. Nunes, R. Marcuzzi, I. Vizzo, J. Behley, and C. Stachniss, “Automatic Labeling to Generate Training Data for Online LiDAR-Based Moving Object Segmentation,” *IEEE Robotics and Automation Letters (RA-L)*, vol. 7, no. 3, pp. 6107–6114, 2022.
- [8] R. Marcuzzi, L. Nunes, L. Wiesmann, I. Vizzo, J. Behley, and C. Stachniss, “Contrastive Instance Association for 4D Panoptic Segmentation using Sequences of 3D LiDAR Scans,” *IEEE Robotics and Automation Letters (RA-L)*, vol. 7, no. 2, pp. 1550–1557, 2022.
- [9] B. Mersch, X. Chen, I. Vizzo, L. Nunes, J. Behley, and C. Stachniss, “Receding Moving Object Segmentation in 3D LiDAR Data Using Sparse 4D Convolutions,” *IEEE Robotics and Automation Letters (RA-L)*, vol. 7, no. 3, pp. 7503–7510, 2022.
- [10] L. Nunes, R. Marcuzzi, X. Chen, J. Behley, and C. Stachniss, “SegContrast: 3D Point Cloud Feature Representation Learning through Self-supervised Segment Discrimination,” *IEEE Robotics and Automation Letters (RA-L)*, vol. 7, no. 2, pp. 2116–2123, 2022.
- [11] L. Nunes, X. Chen, R. Marcuzzi, A. Osep, L. Leal-Taixe, C. Stachniss, and J. Behley, “Unsupervised Class-Agnostic Instance Segmentation of 3D LiDAR Data for Autonomous Vehicles,” *IEEE Robotics and Automation Letters (RA-L)*, 2022.

Peer-reviewed conference papers

- [1] T. Faasch, J. Kall, L. Nunes, J. Behley, and C. Stachniss, “TASE: Truncation-Aware Semantic Embeddings for 3D Scene Understanding and Editing,” in *Proc. of the Europ. Conf. on Computer Vision (ECCV)*, 2026.
- [2] R. Marcuzzi, L. Nunes, E. Marks, X. Zhong, J. Behley, and C. Stachniss, “Vision-Based Panoptic Occupancy Prediction in Urban Environments,” in *Proceedings of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2026.
- [3] Y. Pan, T. Sun, L. Zhu, L. Nunes, I. Armeni, J. Behley, and C. Stachniss, “Register Any Point: Scaling 3D Point Cloud Registration by Flow Matching,” in *Proc. of the Europ. Conf. on Computer Vision (ECCV)*, 2026.
- [4] Y. Chong, L. Nunes, F. Magistri, X. Zhong, J. Behley, and C. Stachniss, “Zero-Shot Semantic Segmentation for Robots in Agriculture,” in *Proceedings of the IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2025.
- [5] E. Marks, L. Nunes, F. Magistri, M. Sodano, R. Marcuzzi, L. Zimmermann, J. Behley, and C. Stachniss, “Tree Skeletonization from 3D Point Clouds by Denoising Diffusion,” in *Proceedings of the IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, 2025.
- [6] L. Nunes, R. Marcuzzi, B. Mersch, J. Behley, and C. Stachniss, “Scaling Diffusion Models to Real-World 3D LiDAR Scene Completion,” in *Proceedings of the IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [7] M. Sodano, F. Magistri, L. Nunes, J. Behley, and C. Stachniss, “Open-World Semantic Segmentation Including Class Similarity,” in *Proceedings of the IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [8] H. Lim, L. Nunes, B. Mersch, X. Chen, J. Behley, H. Myung, and C. Stachniss, “ERASOR2: Instance-Aware Robust 3D Mapping of the Static World in Dynamic Scenes,” in *Proceedings of Robotics: Science and Systems (RSS)*, 2023.
- [9] L. Nunes, L. Wiesmann, R. Marcuzzi, X. Chen, J. Behley, and C. Stachniss, “Temporal Consistent 3D LiDAR Representation Learning for Semantic Perception in Autonomous Driving,” in *Proceedings of the IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2023.
- [10] I. Vizzo, B. Mersch, L. Nunes, L. Wiesmann, T. Guadagnino, and C. Stachniss, “Toward Reproducible Version-Controlled Perception Platforms: Embracing Simplicity in Autonomous Vehicle Dataset Acquisition,” in *Proc. of the Int. Conf. on Intelligent Transportation Systems Workshops*, accepted, 2023.
- [11] D. Bruno, L. P. N. Matias, J. Amaro, F. S. Osorio, and D. Wolf, “Computer Vision System with 2D and 3D Data Fusion for Detection of Possible Auxiliaries Routes in Stretches of Interdicted Roads,” in *Proceedings of the Annual Hawaii Int. Conf. on System Sciences (HICSS)*, 2019, pp. 7372–7381.
- [12] L. P. N. Matias, M. Sons, J. R. Souza, D. F. Wolf, and C. Stiller, “VelGAN: Vectorial Inpainting Generative Adversarial Network for Depth Maps Object Removal,” in *2019 IEEE Intelligent Vehicles Symposium (IV)*, 2019, pp. 310–316.
- [13] L. P. N. Matias, T. C. Santos, D. F. Wolf, and J. R. Souza, “Trajectory Planning for UGV Using Clothoids,” in *Robotics*, F. Santos Osorio and R. Sales Gonalves, Eds., Springer International Publishing, 2016, pp. 281–298.
- [14] L. P. N. Matias, T. C. Santos, D. F. Wolf, and J. R. Souza, “Path Planning and Autonomous Navigation using AMCL and AD,” in *2015 12th Latin American Robotics Symposium and 2015 3rd Brazilian Symposium on Robotics (LARS-SBR)*, 2015, pp. 320–324.

Preprint papers

- [1] K. Knaebel, G. M. Garcia, C. Schmidt, I. Fradlin, L. Nunes, D. de Geus, and B. Leibe, “SurGe: Improved Surface Geometry in Point Maps,” *arXiv Preprint*, vol. arxiv:2605.31577, 2026.
- [2] L. P. N. Matias, J. R. Souza, and D. F. Wolf, “Environment reconstruction on depth images using Generative Adversarial Networks,” *arXiv Preprint*, vol. arxiv:1912.03992, 2019.

Theses

- [1] L. Nunes, “Learning Discriminative Representations and Generative Approaches for Outdoor 3D LiDAR Data,” PhD thesis, University of Bonn, 2026.
- [2] L. Nunes, “Environment reconstruction on disparity images using surface features and Generative Adversarial Networks,” Master’s thesis, University of São Paulo, 2020.